

Hélder Gonçalves

Lead Software Engineer

Software Architecture · Technical Leadership · Reliability · Generative AI

helder@hgoncalves.uk hgoncalves.uk
linkedin.com/in/helderjgoncalves github.com/helderjgoncalves

SUMMARY

Lead Software Engineer with 10+ years building developer tools in React, TypeScript and .NET, used daily by thousands of enterprise developers. Grew with one product from Junior to Lead, taking on wider scope at each step. Leads through architecture and mentoring: measure before fixing, evolve large systems incrementally behind feature flags rather than rewrite them, and hold design to a bar that survives real code. Drives product-wide work in reliability, cloud architecture, design systems and Generative AI.

10+ years	~15% → ~3%	50%+	2 teams
building developer tools	IDE crash ratio	conflicts auto-resolved	12 engineers led

SKILLS

Architecture:	Software architecture, system design, distributed systems integration, REST APIs, embedded Chromium WebView (CEF / CefGlue), design systems, incremental migration
Reliability:	Site Reliability Engineering (SRE), telemetry and observability, crash, hang and freeze metrics, feature flags, dark launches, progressive delivery
Programming:	TypeScript, JavaScript, React, Node.js, C#, .NET, Avalonia, HTML, CSS, SCSS, C, C++
Testing:	Jest, React Testing Library, unit and component testing
Leadership:	Technical leadership, cross-team direction, mentoring, design and code reviews, Product and UX partnership
AI:	Generative AI, LLM integration, agentic AI, AI-assisted developer tools
Cloud & delivery:	AWS, Snowflake, Cloudflare (R2, Workers, WAF, Zero Trust), CI/CD with GitHub Actions, Docker, Git

EXPERIENCE

2023 – Present

Lead Software Engineer

OutSystems

- **Generative AI Integration:** Led the integration of Mentor, OutSystems' LLM-based agentic AI, into ODC Studio: conversational, AI-assisted development in the visual IDE, live since March 2026.
- **Architecture & Mentoring:** Own the architecture of 2 teams (12 engineers) and mentor their engineers; guide other engineering groups through architecture, design and code reviews.
- **Stability & Observability:** Built the telemetry that pinpoints the IDE's high-impact failures and led the fixes: crash ratio from ~15% to ~3% (80% lower). The approach now covers hangs and freezes.
- **SRE & Progressive Delivery:** Introduced Site Reliability Engineering (SRE) practices and dark launches, so the teams ship to production continuously behind feature flags and roll out gradually.
- **Design System Modernization:** Led the 2026 company-wide Design System modernization from design to production: built its infrastructure and set the bar for what could ship, sending designs back to UX when they would not hold up in code.
- **Merge & Conflict Resolution:** Made merge granular, so edits to different parts of the same screen no longer conflict; automatic text merge now resolves over 50% of the rest, in a redesigned merge UI.
- **Desktop Notifications:** Designed and built desktop notifications into the IDE from scratch, giving it a channel to reach developers outside its own window.
- **Cloud Platform Client:** Re-architected the desktop IDE's integration with OutSystems Developer Cloud (ODC), from a single backend dependency to distributed cloud services.
- **Incremental Platform Migration:** With the wider team, made the multi-year move off Windows-only WPF to React and TypeScript in an embedded Chromium WebView, component by component behind feature flags, never forking; it gained macOS.

2020 – 2023 **Senior Software Engineer**
OutSystems

2018 – 2020 **Software Engineer**
OutSystems

2016 – 2018 **Junior Software Engineer**
OutSystems

INTERNSHIPS

2016 **Software Developer Internship**
TU Darmstadt

- Selected for an Erasmus+ exchange at TU Darmstadt's Institute for Scientific Computing. Developed optimised software modules for post-quantum cryptography research, including an Arithmetic Multiple Precision library that outperformed GMP in key operations.
- Parallelised and optimised lattice basis reduction algorithms (LLL and Qiao's Jacobi method) to accelerate cryptographic computations.
- Formed the basis of the MSc thesis at Universidade do Minho.

2015 **Summer Internship in Advanced Computing Research**
UT Austin

- Built a multithreaded C++ framework at UT Austin for the GUIDE project (Graph Unified Irregular Distributed Environment), enabling parallel and distributed operations over complex graph structures.

EDUCATION

2011 – 2016 **Universidade do Minho (University of Minho)**
Master's (MSc) and Bachelor's (BSc) in Informatics Engineering
Thesis: Towards an efficient lattice basis reduction implementation. Coursework: Parallel and Distributed Computing, Distributed Systems & Cryptography, Computer Graphics.

PROJECTS

2026 – Present **Personal Website and Travel Guide (hgoncalves.uk)**
Designed and built the site end to end: UX, visual design, front end and infrastructure. React and TypeScript, pre-rendered to static HTML for SEO and shipped through a GitHub Actions pipeline that tests every build before it deploys. Photos ship as responsive AVIF, and HDR video as AV1 and HEVC, with encoder settings picked by VMAF score. The video is served from Cloudflare R2 behind cache rules and WAF rules that block hotlinking. A scheduled Cloudflare Worker trips a kill switch before storage reads leave the free tier.
STACK React · TypeScript · Vite · GitHub Actions · Cloudflare R2 · Cloudflare Workers · WAF · AV1 / HEVC

2023 – Present **Self-Hosted Home Lab (hgoncalves.uk/homelab)**
NAS-based private cloud serving the whole family — photos, files and media (Immich, OpenCloud, Plex), running in Docker. No inbound ports: traffic arrives through Cloudflare Tunnel and a Traefik reverse proxy, with zero-trust access at the edge (OAuth, mTLS, service tokens). It also runs DNS over HTTPS and layered monitoring with alerting. The whole stack is declared in Git, so it redeploys onto fresh hardware.
STACK Docker · Cloudflare Zero Trust · Traefik · mTLS · OAuth · DNS over HTTPS · Configuration-as-Code

CERTIFICATIONS & AWARDS

CLF-C01 AWS Certified Cloud Practitioner, Amazon Web Services
CSD Certified SCRUM Developer, Scopphu
2023 **Top Performer Award**, OutSystems: Company-wide recognition for delivering high-impact technical results.

LANGUAGES

Portuguese (native), English (fluent)